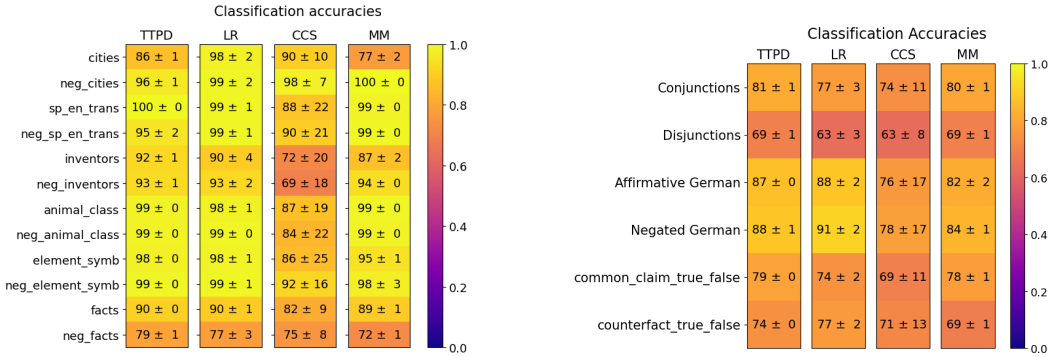

(a) (b)

Figure 6: Generalization accuracies of TTPD, LR, CCS and MM. Mean and standard deviation computed from 20 training runs, each on a different random sample of the training data.

In step (i), we leverage the insight from the previous sections that different types of true and false statements separate well along $\mathbf{t}_G$. However, statements with different polarities need slightly different biases for accurate classification (see Figure 1). To accommodate this, we learn the polarity direction $\mathbf{p}$ in step (ii). To classify a new statement, TTPD projects its activation vector onto $\mathbf{t}_G$ and $\mathbf{p}$ and applies the trained Logistic Regression classifier in the resulting 2D space to predict the truth label.

We benchmark TTPD against three widely used approaches that represent the current state-of-the-art: (i) Logistic Regression (LR): Used by Burns et al. [2023] and Marks and Tegmark [2023] to classify statements as true or false based on internal model activations and by Li et al. [2024] to find truthful directions. (ii) Contrast Consistent Search (CCS) by Burns et al. [2023]: A method that identifies a direction satisfying logical consistency properties given contrast pairs of statements with opposite truth values. We create contrast pairs by pairing each affirmative statement with its negated counterpart, as done in Marks and Tegmark [2023]. (iii) Mass Mean (MM) probe by Marks and Tegmark [2023]: This method derives a truth direction $\mathbf{t}_{\text{MM}}$ by calculating the difference between the mean of all true statements $\boldsymbol{\mu}^+$ and the mean of all false statements $\boldsymbol{\mu}^-$, such that $\mathbf{t}_{\text{MM}} = \boldsymbol{\mu}^+ - \boldsymbol{\mu}^-$. To ensure a fair comparison, we have extended the MM probe by incorporating a learned bias term. This bias is learned by fitting a LR classifier to the one-dimensional projections $\mathbf{a}^\top \mathbf{t}_{\text{MM}}$.

5.1 Unseen topics and statement types

Figure 6a shows the generalisation accuracy of the classifiers to unseen topics. We trained the classifiers on an equal number of activations from all but one topic-specific dataset (affirmative and negated version), holding out this excluded dataset for testing. TTPD and LR generalize similarly well, achieving average accuracies of $93.9 \pm 0.2\%$ and $94.6 \pm 0.7\%$, respectively, compared to $84.8 \pm 6.4\%$ for CCS and $92.2 \pm 0.4\%$ for MM.

Next, we evaluate the classifiers' generalization to unseen statement types, training solely on activations from English affirmative and negated statements. Figure 6b displays classification accuracies for logical conjunctions, disjunctions, and German translations of affirmative and negated statements, averaged across multiple datasets. Individual dataset accuracies are presented in Figure 9 of Appendix E. TTPD outperforms LR and CCS in generalizing to logical conjunctions and disjunctions. It also exhibits impressive classification accuracies on German statements, only a few percentage points lower than their English counterparts. For the more diverse and occasionally ambiguous test sets `common_claim_true_false` and `counterfact_true_false`, which closely resemble the training data in form, TTPD and LR perform similarly well.

5.2 Real-world lies

The ultimate goal of our research extends beyond distinguishing simple true and false statements. We aim to reliably detect sophisticated lies generated by LLMs [Scheurer et al., 2024, Hubinger et al., 2024, Järviniemi and Hubinger, 2024]. In such settings we cannot trust the LLM outputs as a